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$$z^4 + 4$$

$$\text{stel: } z^2 = t$$

$$\text{dan: } z^4 + 4 = t^2 + 4$$

$$t^2 = -4$$

$$t_{1,2} = \pm 2i$$

$$t_1 = 2i \Rightarrow z_{1,2}^2 = 2i$$

$$z_1 = 1 + i$$

$$z_2 = -1 - i$$

$$t_2 = -2i \Rightarrow z_{3,4}^2 = -2i$$

$$z_3 = 1 + i$$

$$z_4 = -1 + i$$

$$\text{dus: } z^4 + 4 = (z - (1 + i))(z - (1 - i))(z - (-1 - i))(z - (-1 + i))$$

$$z^4 + 4 = (z^2 - 2z + 2)(z^2 + 2z + 2)$$

References